



Baffle optimization design and mixing performance study of asymmetric tesla vortex micromixer

Pu Liu, Yiheng Yue, Yu Guo, Wei Yang, Lili Zang^{*}

School of Electronic Engineering and Intelligent Manufacturing, Anqing Normal University, Anqing 246133, PR China

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ABSTRACT

Aiming at the bottleneck problem of insufficient mixing efficiency of traditional micromixers under low Reynolds number conditions, this paper innovatively proposes an asymmetric Tesla vortex micromixer (ATVM) with baffles. This paper systematically studies its mixing mechanism and performance by constructing a multi-physics field coupling numerical model. The study focuses on the influence of baffle structure (no baffle, rectangular, triangular and square), baffle height (20 μm , 50 μm and 75 μm) and number (4 and 6) on mixing efficiency, pressure drop and energy consumption. The results show that the ATVM with 6 square baffles (the baffle height of 50 μm) has the best comprehensive performance and lowest energy consumption with a mixing efficiency of up to 99.42 % at $Re = 100$. The mixing mechanism analysis shows that the Tesla valve structure generates vortices through spontaneous flow separation, breaking the laminar state. The baffle structure further enhances chaotic convection and significantly improves the mixing efficiency. This study not only deepens the understanding of microscale mixing mechanisms, but also provides a new paradigm of high efficiency and low consumption for the application of microfluidic technology in the fields of biochemical analysis and drug synthesis.

1. Introduction

The rapid development of microfluidics is profoundly reshaping the scientific paradigm and technical boundaries in the fields of biochemical analysis, drug delivery and point-of-care diagnosis [1–4]. As the core platform for precise manipulation of microfluidics, microfluidics systems use the hydrodynamic effects of centimeter-scale flow channels to upgrade the internal surface [5,6]. Microfluidic chips successfully compress the volume of the reaction system to the microliter level, showing revolutionary advantages in improving reaction rates and reducing reagent consumption [7,8]. In this technology system, micromixers are key units that determine system performance [9,10]. The mixing efficiency of micromixers directly affects core indicators such as biochemical detection sensitivity, drug synthesis purity and point-of-care diagnosis accuracy [11,12]. However, under low Reynolds number ($Re < 100$) flow states, the laminar-dominated diffusion mechanism causes traditional micromixers to generally face the bottleneck problem of insufficient mass transfer efficiency [13,14]. This physical limitation has become the main obstacle to the expansion of microfluidics technology to high-throughput and high-precision application scenarios [15,16]. The current mainstream passive mixing strategy relies on the

chaotic convection effect induced by the geometric topology of the flow channel, such as classic designs such as serpentine channels and fish-bone structures, which achieve mixing enhancement by enhancing streamline folding and interface stretching [17,18]. However, such structures are susceptible to viscous dissipation effects under low Reynolds number conditions, making it difficult to form effective secondary flow disturbances, resulting in limited improvement in mixing efficiency [19–22]. Although active mixing technology can break through diffusion limitations through external field excitations such as sound, electricity, and magnetism [23–25]. However, its complex drive system and high energy consumption cost significantly increase the difficulty of equipment integration, which is in essential conflict with the core demands of miniaturization and portability of microfluidic technology [26,27].

In recent years, bionic fluid dynamics has provided a new research direction for breaking through the bottleneck of microscale mixing efficiency [28]. The biological mass transfer system formed by millions of years of evolutionary optimization in nature, such as the vortex capture mechanism of the lateral line organ of fish and the fractal transport network of plant roots, exhibits high-efficiency and low-consumption mass transfer characteristics [29,30]. Inspired by this, the Tesla valve

^{*} Corresponding author.

E-mail addresses: yangwei13@aqnu.edu.cn (W. Yang), zangzang55555@163.com (L. Zang).

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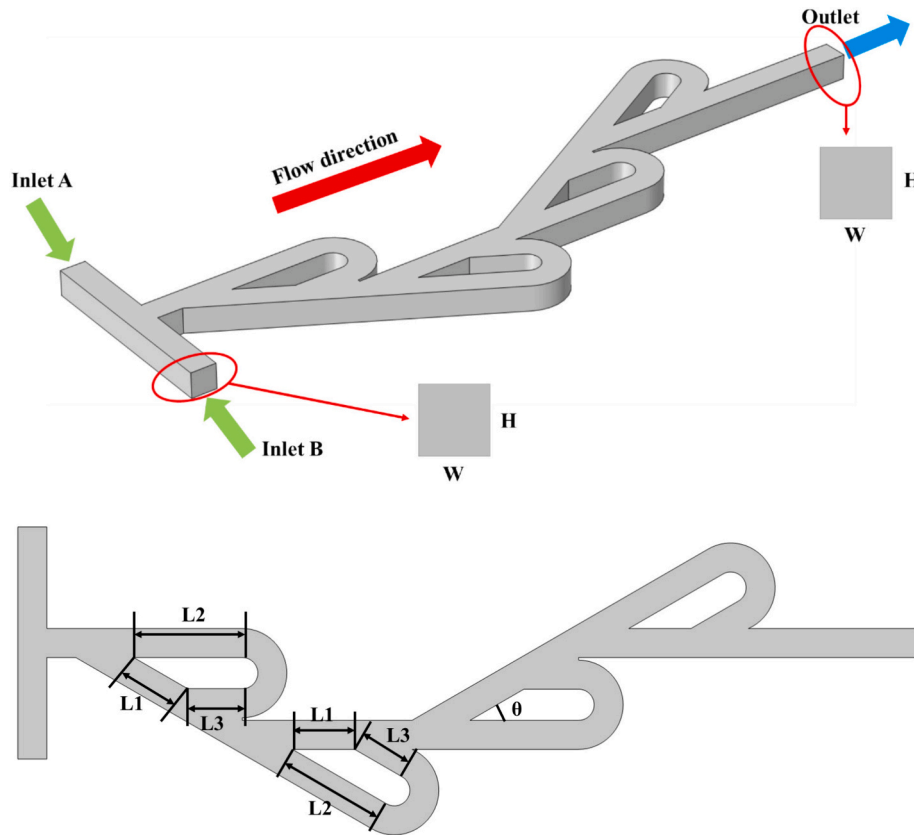


Fig. 1. Schematic diagram of the structural dimensions of the basic model of an asymmetric Tesla vortex micromixer (BM-ATVM).

structure with directional flow resistance characteristics has gradually become an emerging hotspot in micromixer research due to its unique asymmetric flow channel design and spontaneous vortex generation ability [31]. The classic Tesla valve achieves valveless unidirectional flow control at the macroscale through the periodic arrangement of contraction-expansion units, and its inherent fluid adaptability shows significant advantages in microscale environments. Asymmetric geometric structures can induce spontaneous flow instability under low Reynolds number conditions and break the laminar boundary layer through directional vortex generation [32]. The dynamic coupling of the contraction section and the expansion section can produce periodic pressure disturbances and enhance the lateral migration of fluid particles [33]. Le et al. developed a microfluidic Tesla mixer with integrated three-dimensional obstacles [34]. Through computational fluid dynamics simulation and experimental verification, the conical obstacles significantly enhanced the mixing efficiency. The mixer can achieve precise control of the size and monodispersity of polylactic-co-glycolic acid (PLGA) nanoparticles by regulating the solvent type, polymer concentration/molecular weight and flow rate conditions, and the uniformity of nanoparticles prepared by the traditional batch method is improved by 3 times. Sun et al. introduced a three-dimensional fishbone structure into a Tesla micromixer to enhance the lateral flow and vortex of the fluid [35]. They showed that the concave fishbone structure increased the mixing efficiency by 65 % at $Re = 5$, while the convex structure had the best energy consumption ratio. The limitation of their research is that they did not systematically optimize the geometric parameters of the fishbone structure (such as angle, spacing) and the quantitative effect of its spatial arrangement on the mixing performance. Cui et al. optimized the passive micromixer structure based on a three-dimensional Tesla valve through numerical simulation [36]. They focused on its influence on the mixing performance of the functional gradient material of Fe₃O₄ particles/epoxy resin E51 matrix and determined the optimal combination of five key parameters. The

limitation of their study is that they did not systematically analyze the quantitative influence of geometric structure (such as obstacle shape and position) on mixing efficiency. Li et al. designed an asymmetrically arranged Tesla micromixer and systematically studied the effects of obstacle arrangement, geometry, and size on mixing performance by embedding an obstacle optimization structure [37]. The limitation of their research is that the obstacle optimization only stays at the parameter adjustment stage and lacks a full exploration of the number and spatial arrangement of obstacles. Buglie et al. experimentally characterized the fluid mixing performance of a two-dimensional reverse multistage Tesla micromixer at different Reynolds numbers and valve stages [38]. They used a simplified design to reduce the difficulty of manufacturing and found that the mixing performance increased steadily with the number of valve stages in the low Reynolds number ($Re \leq 20$) region, while the first valve achieved significant enhancement in the high Reynolds number ($Re > 40$) region. The limitations of this study include the limited experimental Reynolds number range and large measurement uncertainty, which may lead to micro-channel size deviations and affect practical scalability. However, its inherent vortex evolution mechanism and geometric parameter sensitivity have not been systematically analyzed [39]. In particular, there is still a theoretical gap in how to achieve coordinated regulation of vortex intensity and energy loss through local structural optimization [40]. The low Reynolds number range is $Re < 100$. Its core importance is reflected in the following three aspects. In microfluidic biochemical analysis scenarios (such as organ chips, single-cell manipulation), the fluid inertial force is significantly weaker than the viscous force ($Re \ll 1$), resulting in the laminar-dominated mass transfer process being highly dependent on slow molecular diffusion, and the efficiency of traditional mixers is generally less than 70 %. This design uses the asymmetric structure of the Tesla valve to spontaneously induce flow field instability at low Re , combined with the directional secondary vortex generated by the baffle, to transform the pure diffusion mechanism into a convection-

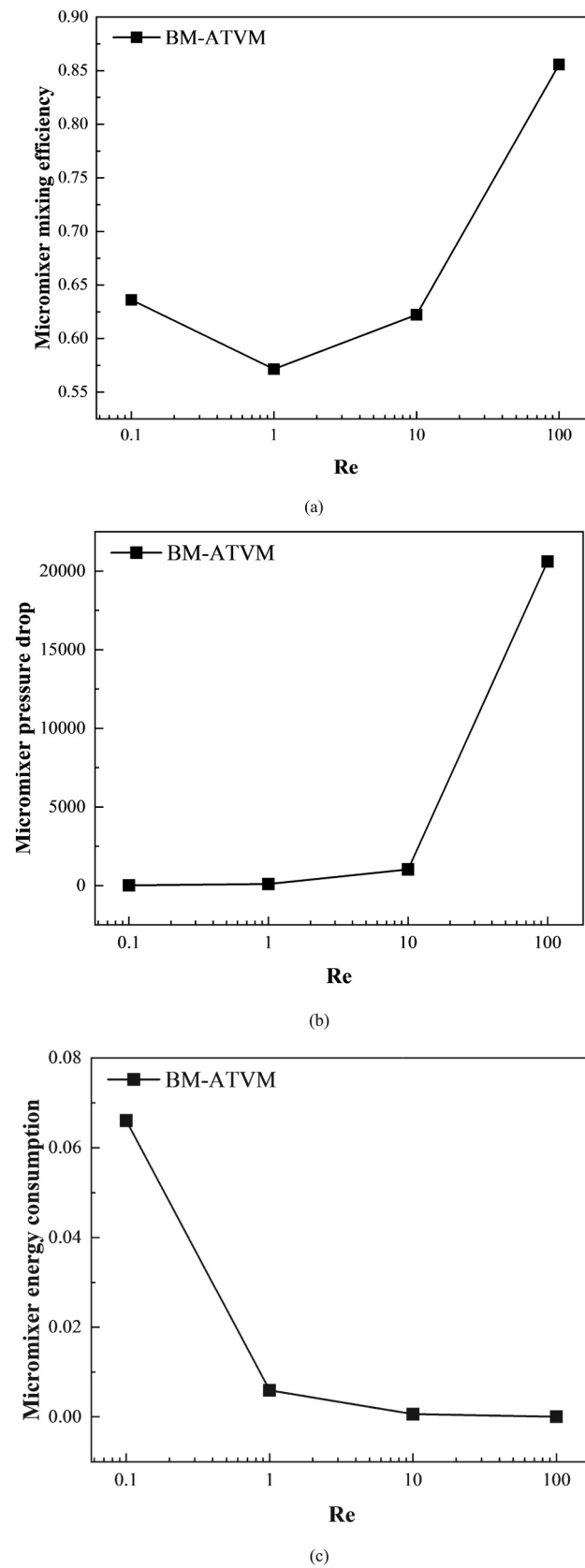


Fig. 2. Mixing efficiency curve, pressure drop curve and mixing energy consumption curve of BM-ATVM at different Re (a) Mixing efficiency (b) Pressure drop (c) Mixing energy consumption.

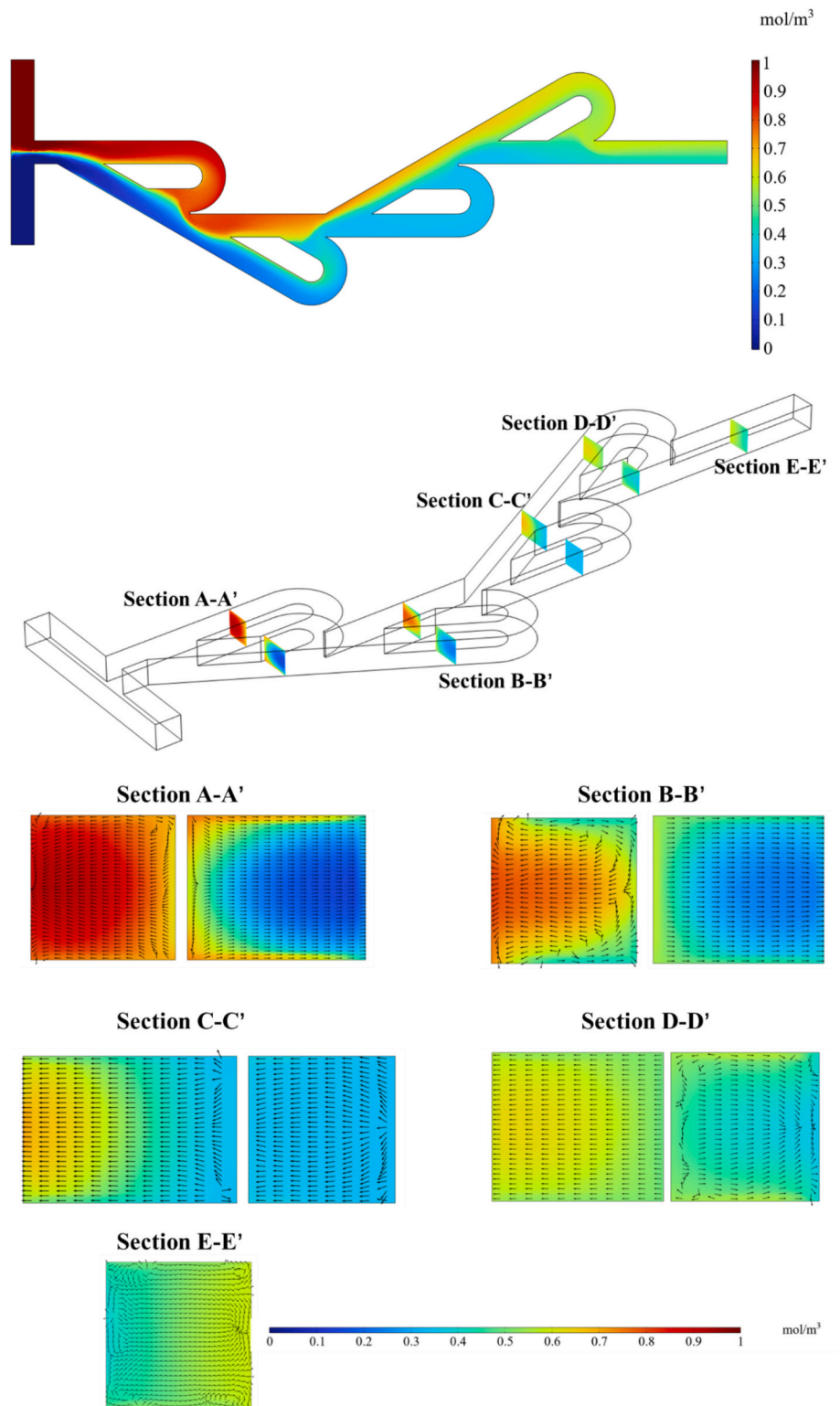


Fig. 3. Schematic diagram of concentration distribution in horizontal section, concentration distribution in vertical section and velocity arrow distribution of BM-ATVM at $Re = 100$.

diffusion synergistic mechanism, so that the mixing efficiency is significantly improved at $Re = 100$. Compared with high- Re turbulent mixing, low- Re operation can avoid shear denaturation of biomacromolecules and reduce system energy consumption. This characteristic of achieving efficient mixing under laminar flow is the core advantage of microfluidic technology in fields such as POCT diagnosis and nanodrug synthesis.

In response to the above scientific problems and technical challenges, this study innovatively proposed an asymmetric Tesla Vortex Micromixer (ATVM) with an integrated baffle structure. Through multi-physics field coupling modeling and parametric design, a breakthrough improvement in mixing performance is achieved. This paper breaks through the symmetrical flow channel layout of the traditional Tesla

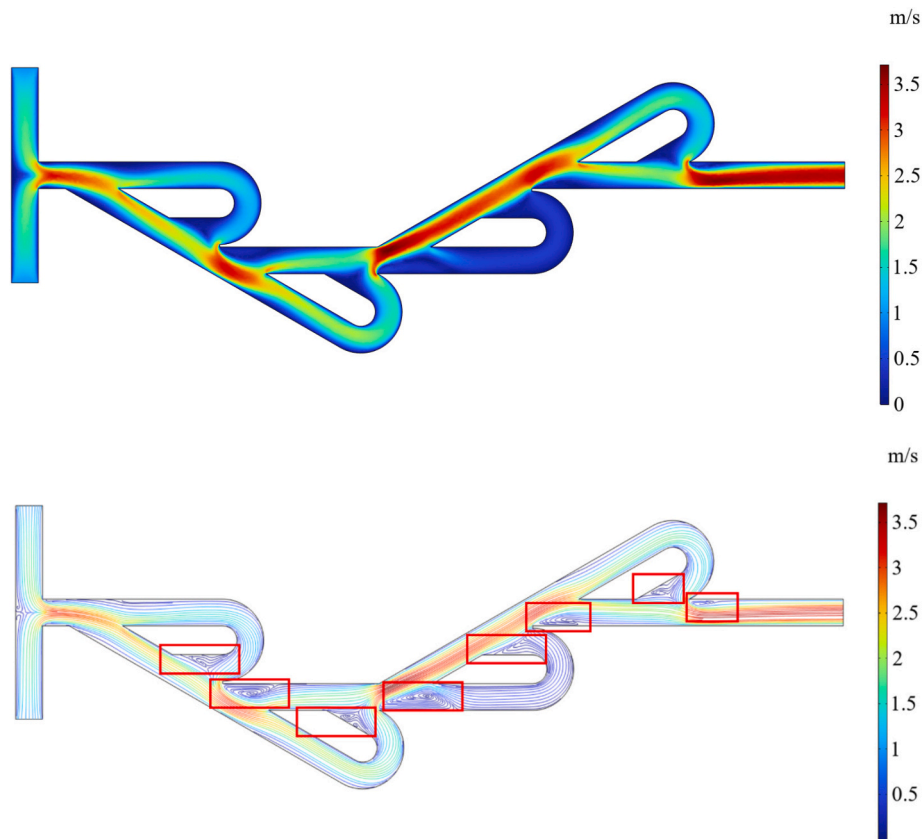


Fig. 4. Velocity streamline distribution at the horizontal section of BM-ATVM at $Re = 100$.

valve and introduces the synergistic mechanism of the asymmetric contraction-expansion unit and the three-dimensional baffle structure. By optimizing the geometric parameters of the baffle, a composite flow field with multiple vortex excitation characteristics is constructed, and the streamline folding degree and interface deformation rate are significantly enhanced under low Reynolds number conditions. This paper establishes a multi-physics field coupling model covering the Navier-Stokes equation and the convection-diffusion equation, and systematically reveals the influence of the baffle structure on the vortex generation mode, evolution path and energy transfer efficiency. The mixing efficiency of the optimal configuration ATVM reaches 99.42 % at $Re = 100$, which is 13.85 % higher than the baseline model, and the unit mixing efficiency pressure drop is reduced by 21.3 %, successfully achieving the coordinated optimization of efficient mixing and low energy consumption.

2. Numerical simulation methodology

2.1. Physical model construction

Tesla Valve, as a one-way flow structure without moving parts, has shown unique application potential in the field of microfluidics in recent years. Especially in the design of micromixers, its asymmetric geometric characteristics and passive flow control mechanism provide innovative solutions for improving mixing efficiency. Tesla valve induces flow direction-dependent resistance difference at low Reynolds number through asymmetric contraction-expansion flow channel design. During forward flow, the gradually expanding geometry of the flow channel accelerates the fluid and reduces the local pressure drop. During reverse flow, the sudden expansion of the cross section causes flow separation and vortex generation, which significantly increases the resistance. Compared with traditional chaotic flow channels (such as serpentine or

herringbone channels), the Tesla valve structure spontaneously generates disturbances through flow separation without relying on complex geometry or external energy input. The forward low resistance characteristics of the Tesla valve can partially offset the increase in pressure drop caused by structural complexity. A large number of studies have shown that its unit mixing efficiency energy consumption is better than that of traditional structures. Based on the Tesla valve structure, this paper designs a novel asymmetric Tesla vortex micromixer. Among them, the structure schematic diagram of basic model of the asymmetric Tesla vortex micromixer (BM-ATVM) is shown in Fig. 1. The cross-sectional dimensions of the microchannel inlet and outlet are $W \times H = 100 \mu\text{m} \times 100 \mu\text{m}$. The mixing area dimensions of the Tesla valve structure are $L1 = 375 \mu\text{m}$, $L2 = 188 \mu\text{m}$, and $L3 = 215 \mu\text{m}$, respectively. The angle of the Tesla valve structure is $\theta = 30^\circ$.

2.2. Boundary condition settings

In the numerical simulation study of the micromixer, the establishment of the model relies on a series of physical and mathematical assumptions to ensure the controllability of the calculation and the physical rationality of the results. The fluid dynamics part is based on the assumption of incompressible Newtonian fluid, that is, the fluid density is constant and the shear stress is linearly related to the strain rate. It is assumed that the solute concentration is much lower than the solvent, thereby ignoring the effect of concentration changes on fluid density and viscosity. The inlet A and inlet B of the micromixer use velocity inlets ($u_{in} = 0.001\text{--}1 \text{ m/s}$). The outlet adopts free outflow conditions ($P_{out} = 0 \text{ Pa}$) to ensure mass conservation. The wall treatment selects no slip ($u_{wall} = 0 \text{ m/s}$) based on the surface characteristics of the PDMS material. In the material transport boundary, the inlet concentration is set to a constant value. The concentration of inlet A is $c_A = 1 \text{ mol/m}^3$. The concentration of inlet B is $c_B = 0 \text{ mol/m}^3$. The outlet adopts

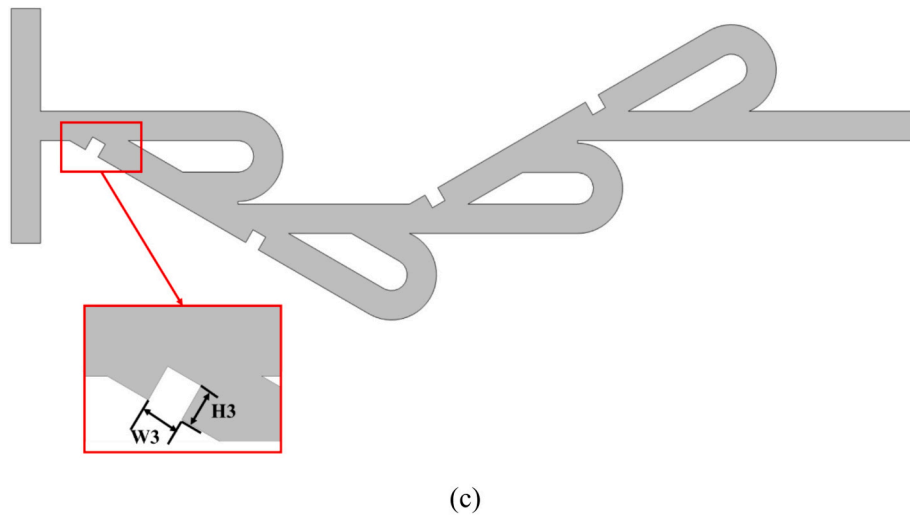
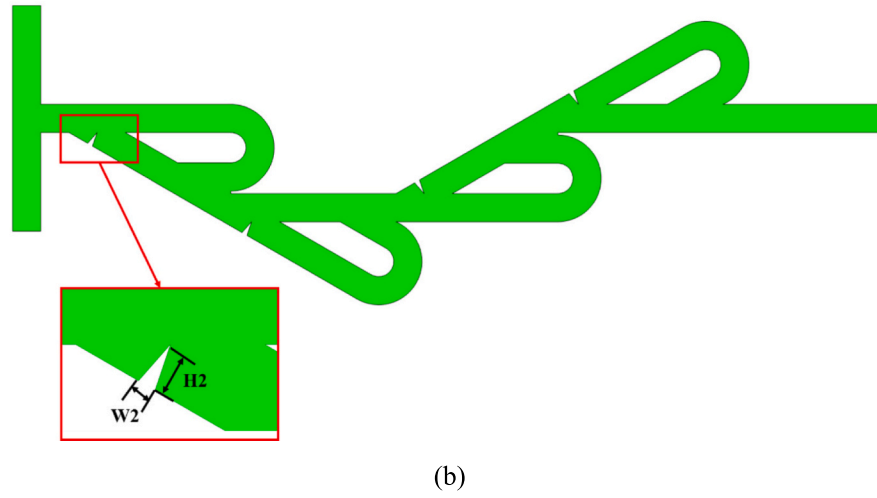
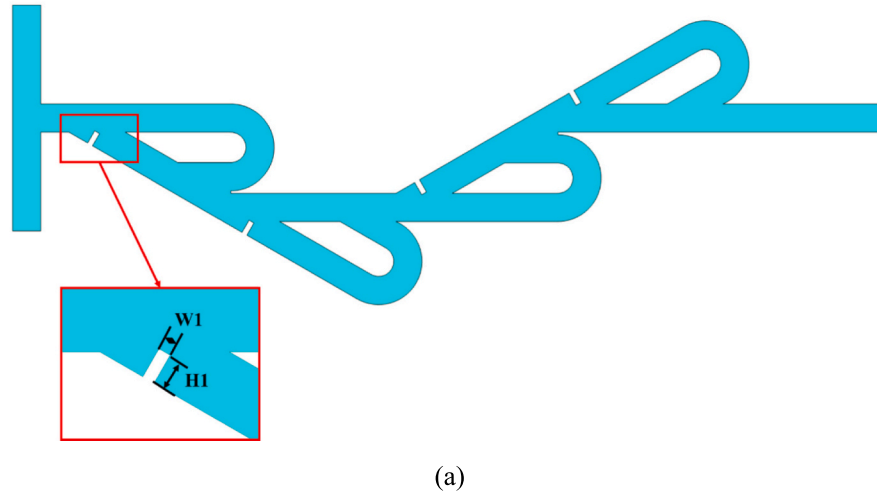


Fig. 5. Schematic diagram of the structure of BM-ATVM with different structural baffles (a) rectangular baffle, RB-ATVM ($w_1 = 50 \mu\text{m}$, $H_1 = 100 \mu\text{m}$) (b) triangular baffle, TB-ATVM ($w_2 = 50 \mu\text{m}$, $H_2 = 100 \mu\text{m}$) (c) square baffle, SB-ATVM ($w_3 = 100 \mu\text{m}$, $H_1 = 100 \mu\text{m}$).

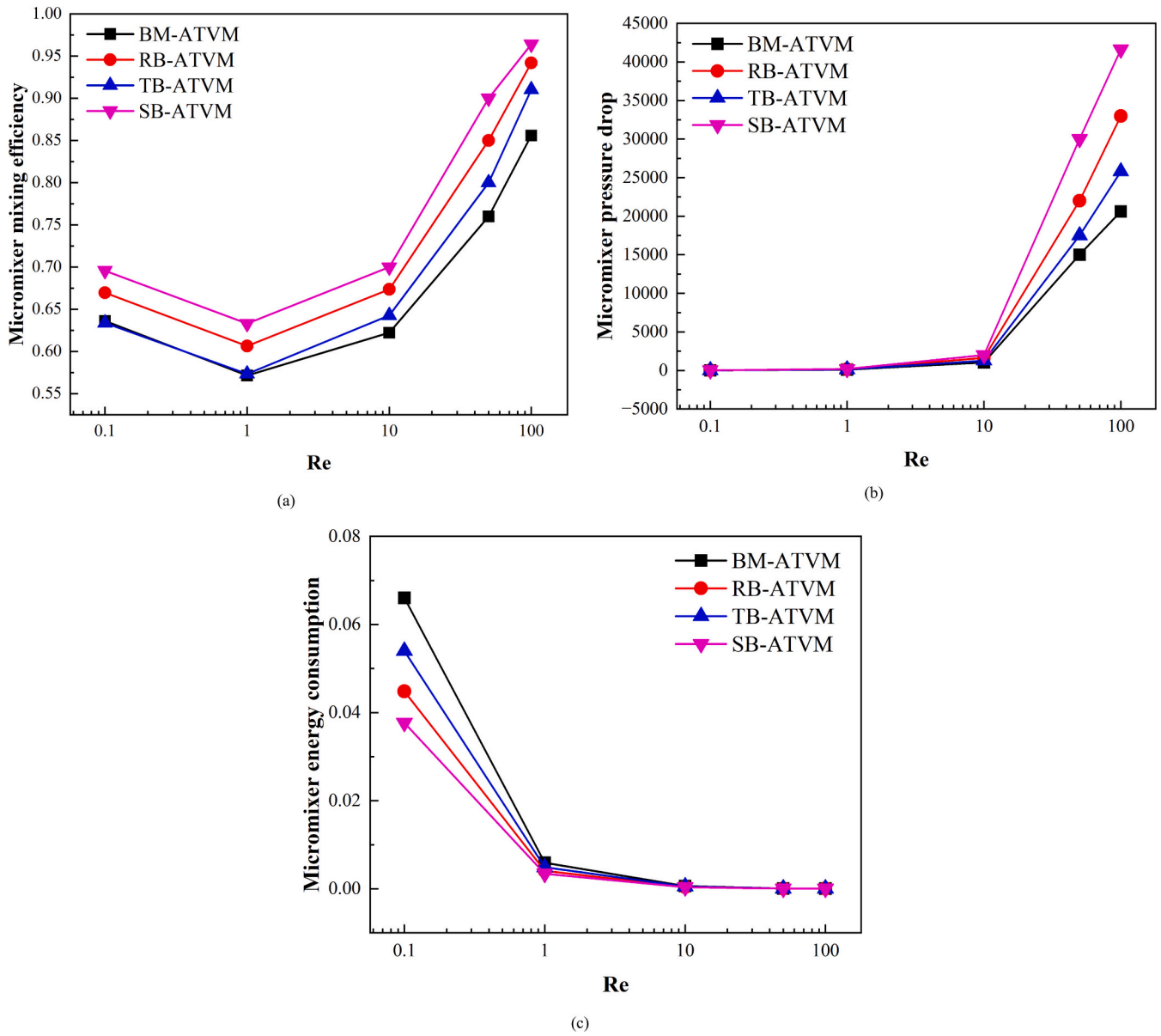


Fig. 6. Mixing efficiency curve, pressure drop curve and mixing energy consumption curve of RB-ATVM, TB-ATVM and SB-ATVM at different Re (a) Mixing efficiency (b) Pressure drop (c) Mixing energy consumption.

convection-diffusion equilibrium conditions. The governing equations used in the study are shown in S1.1. Through the grid independence test (S1.2) in Fig. S1, the optimal numerical simulation grid structure is obtained, as shown in Fig. S2.

3. Results and discussion

3.1. BM-ATVM mixing mechanism and mixing performance analysis

In this section, the mixing process and mixing mechanism of BM-ATVM at different Reynolds numbers are studied by numerical simulation. As shown in Fig. 2, we plot the micromixer mixing efficiency curve, mixer pressure drop curve and mixer energy consumption curve of BM-ATVM at different Reynolds numbers. As can be seen from Fig. 2 (a), when $Re = 0.1$, the mixing efficiency of the micromixer is very low, about 65 %. When the Reynolds number increases from 0.1 to 10, the mixing efficiency shows a trend of first decreasing and then increasing. However, when $Re = 10$, the mixing efficiency of the micromixer does not increase significantly. When $Re = 100$, the mixing efficiency of the

micromixer increases sharply to 85.57 %. As can be seen from Fig. 2 (b), when $Re = 0.1-10$, the pressure drop of the micromixer does not change significantly. However, when Re increases from 10 to 100, the pressure drop of the micromixer increases significantly to 20,610.50 Pa. In order to more accurately evaluate the mixing performance of the micromixer, the trend of the energy consumption of the micromixer can be seen in Fig. 2 (c). As can be seen from Fig. 2 (c), the rate of change of the energy consumption of the micromixer shows a trend of gradually decreasing with the increase of Re . This is because when $Re = 100$, the mixing efficiency of the micromixer is very high, thereby reducing the mixing energy consumption of the micromixer.

In order to more comprehensively analyze the mixing process and mixing mechanism of BM-ATVM, as shown in Fig. 3, when $Re = 100$, the concentration map at the horizontal section of the main channel of the micromixer, the concentration distribution map on the vertical section of the five main channels and the velocity arrow distribution map were obtained. As can be seen from Fig. 3, at the A-A' section on both sides of the Tesla valve obstacle structure, the concentration on both sides is quite different. With the effect of the separation and confluence

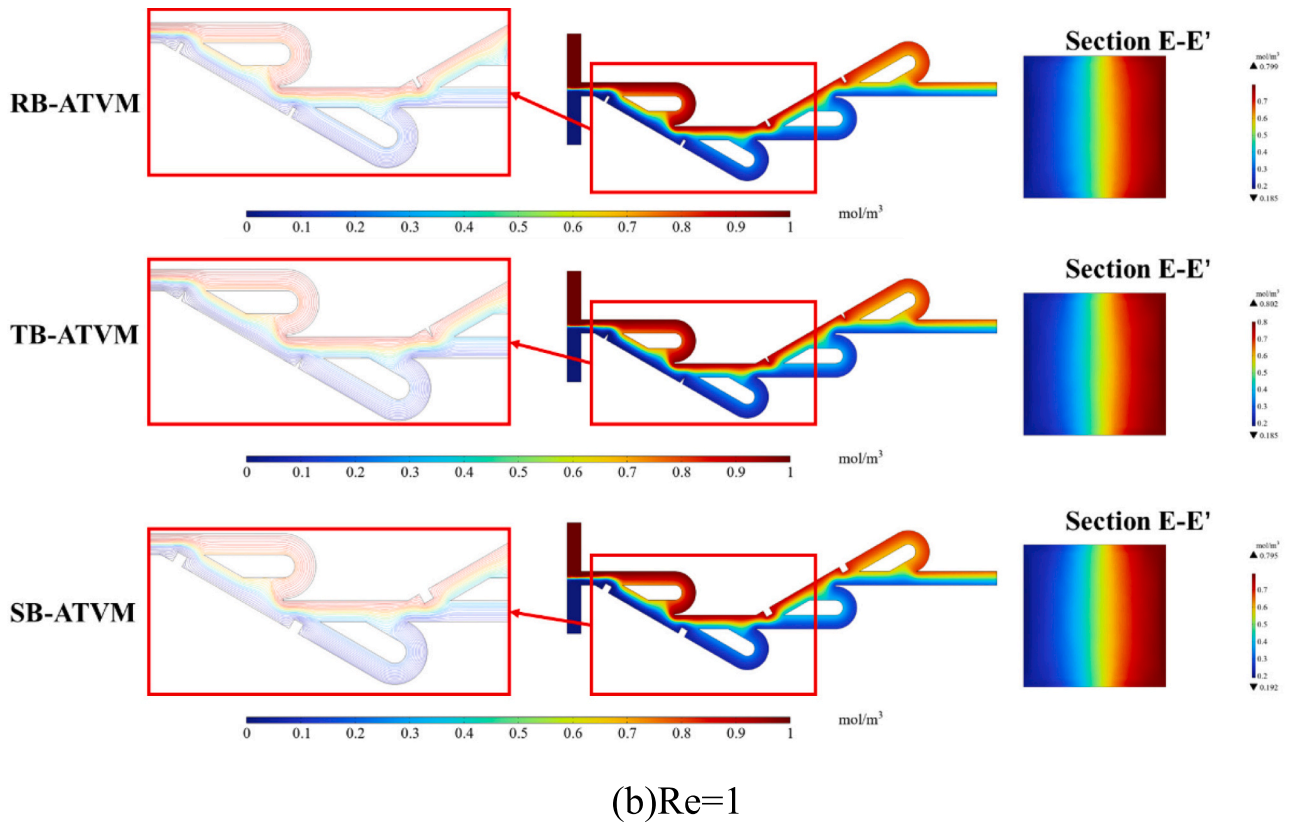
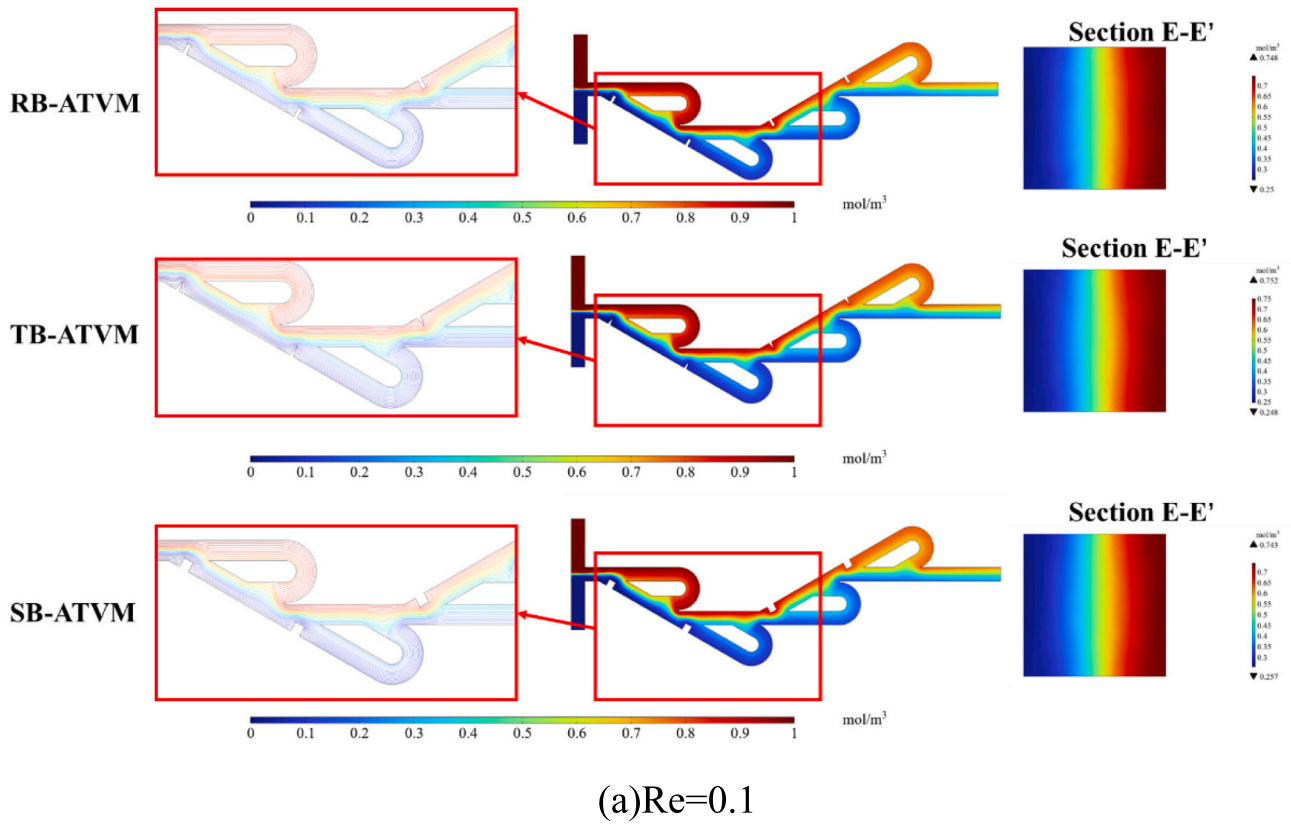


Fig. 7. Concentration field distribution and streamline distribution at the horizontal cross section, outlet concentration distribution and outlet arrow distribution of RB-ATVM, TB-ATVM and SB-ATVM at different Re (a) $Re = 0.1$ (b) $Re = 1$ (c) $Re = 10$ (d) $Re = 100$.

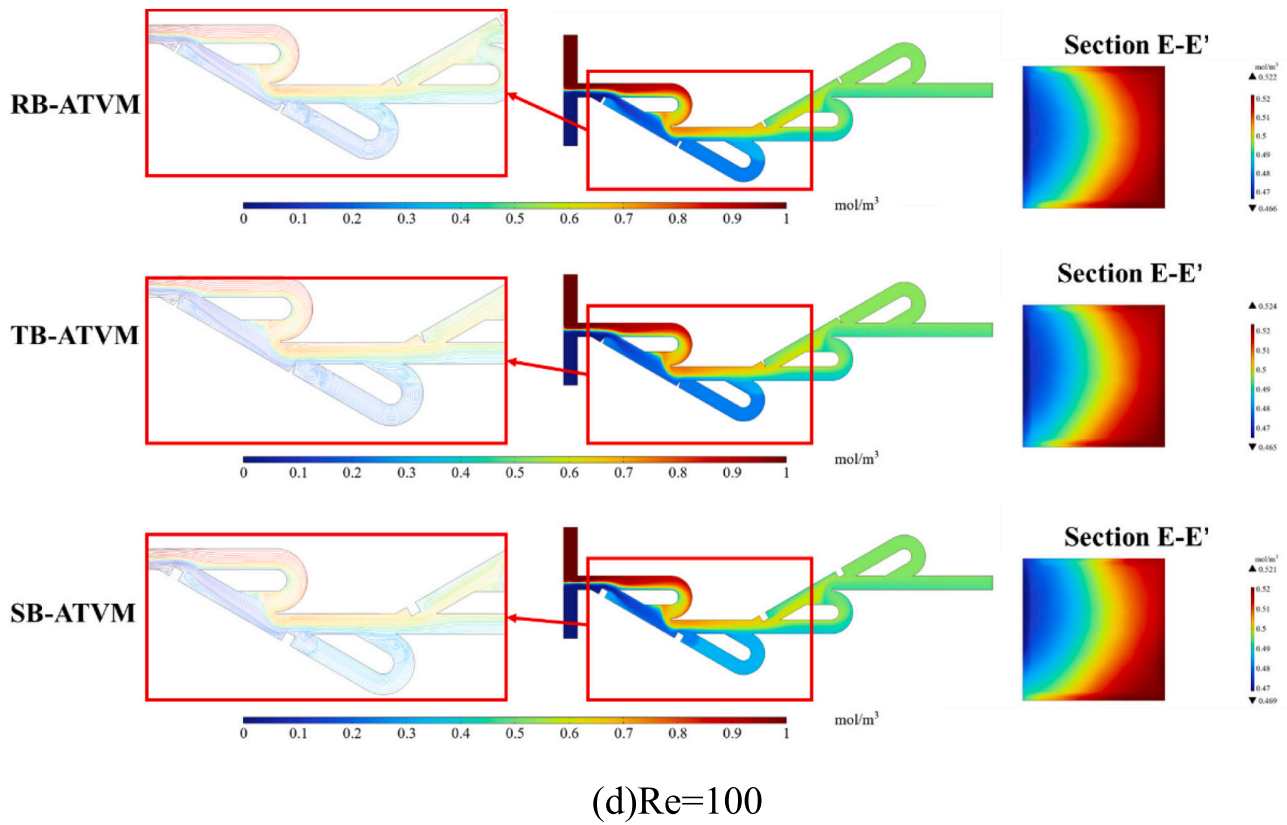
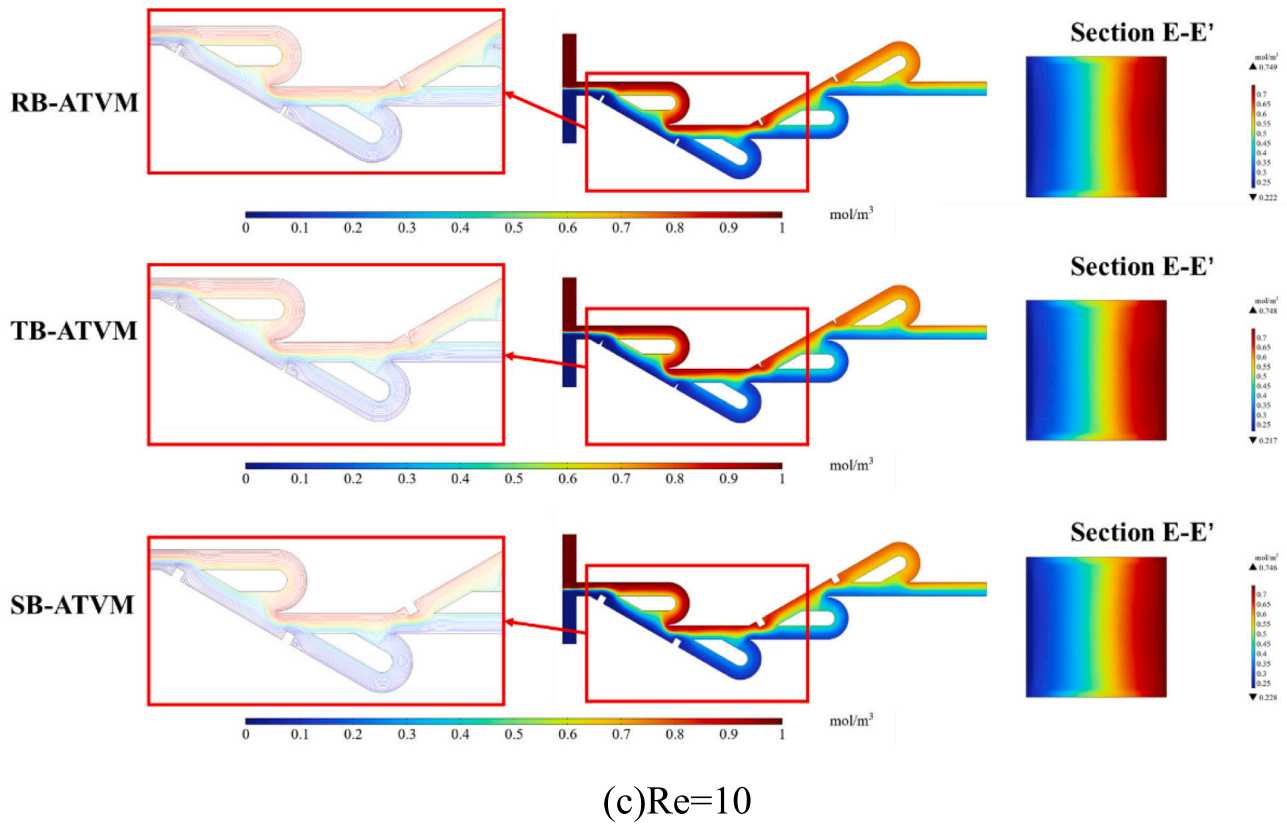


Fig. 7. (continued).

structure of the Tesla valve obstacle, the concentration on both sides of the obstacle gradually approaches. After mixing through multiple mixing units, the fluid concentration is well mixed at the $E-E'$ section. It can

be seen from the velocity arrow distribution map that after passing through the Tesla valve obstacle, the fluid generates multiple velocity vortices, causing fluid disturbance. As a result, the fluid breaks the

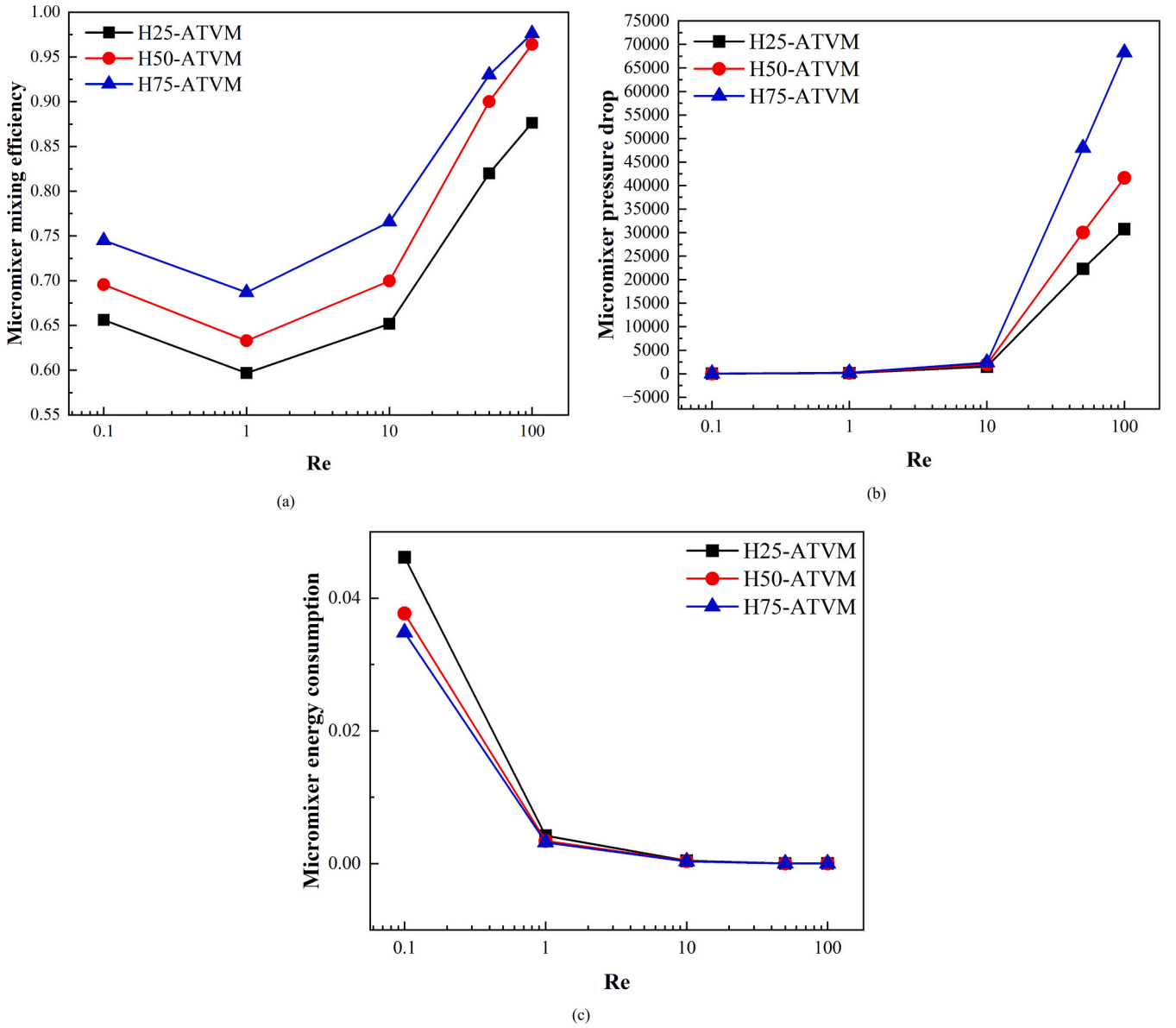


Fig. 8. Mixing efficiency curve, pressure drop curve and mixing energy consumption curve of H25-ATVM, H50-ATVM and H75-ATVM at different Re (a) Mixing efficiency (b) Pressure drop (c) Mixing energy consumption.

laminar flow state at the vertical section and generates chaotic vortices to accelerate mixing. In order to better analyze the situation of generating vortex state in the mixer, as shown in Fig. 4, we obtained the velocity distribution map and velocity streamline distribution map at the horizontal section of the main channel of the micromixer. As can be seen from Fig. 4, the velocity is very large before and after the Tesla valve obstacle. However, the velocity is lower at the curved channel flowing through the Tesla valve obstacle. And multiple vortices are generated at the Tesla valve obstacle. These generated vortices cause the fluid to fold and stretch in the horizontal direction. As a result, the fluid collides sharply, accelerating the mixing of the fluid.

The Tesla-type micromixing device achieves efficient mixing of heterogeneous fluids through its unique flow channel structure design. The periodically arranged separation and combination units in the flow channel will trigger the dynamic reorganization process of the fluid. In the step-by-step transmission of the fluid, the geometric contraction effect of the main channel will cause the two-phase flow section to produce compression deformation. At this time, the shear stress on the fluid unit at the narrow diameter increases. This mechanical energy

conversion significantly enhances the kinetic energy exchange of the fluid micro-groups. As the tributaries after the diversion collide in opposite directions at the confluence node, vortices will form in the flow field. This vortex phenomenon is particularly obvious when the Reynolds number $Re = 100$. The vortices generated by the collision will induce lateral secondary flow in the fluid. This enhanced mixing mechanism is essentially to convert the pressure energy of the fluid into interfacial free energy. By increasing the contact area and enhancing the local concentration gradient, the Fick diffusion and Taylor dispersion effects work synergistically, and finally a rapid homogenization process at the molecular scale is achieved.

3.2. Effects of baffle structures on mixing performance of asymmetric tesla vortex micromixer

According to the research in the previous section, we can get that the mixing efficiency of BM-ATVM is only 85.57 %, which does not reach the complete mixing state. Usually, in order to improve the mixing efficiency, most studies will use the addition of mixing units or the

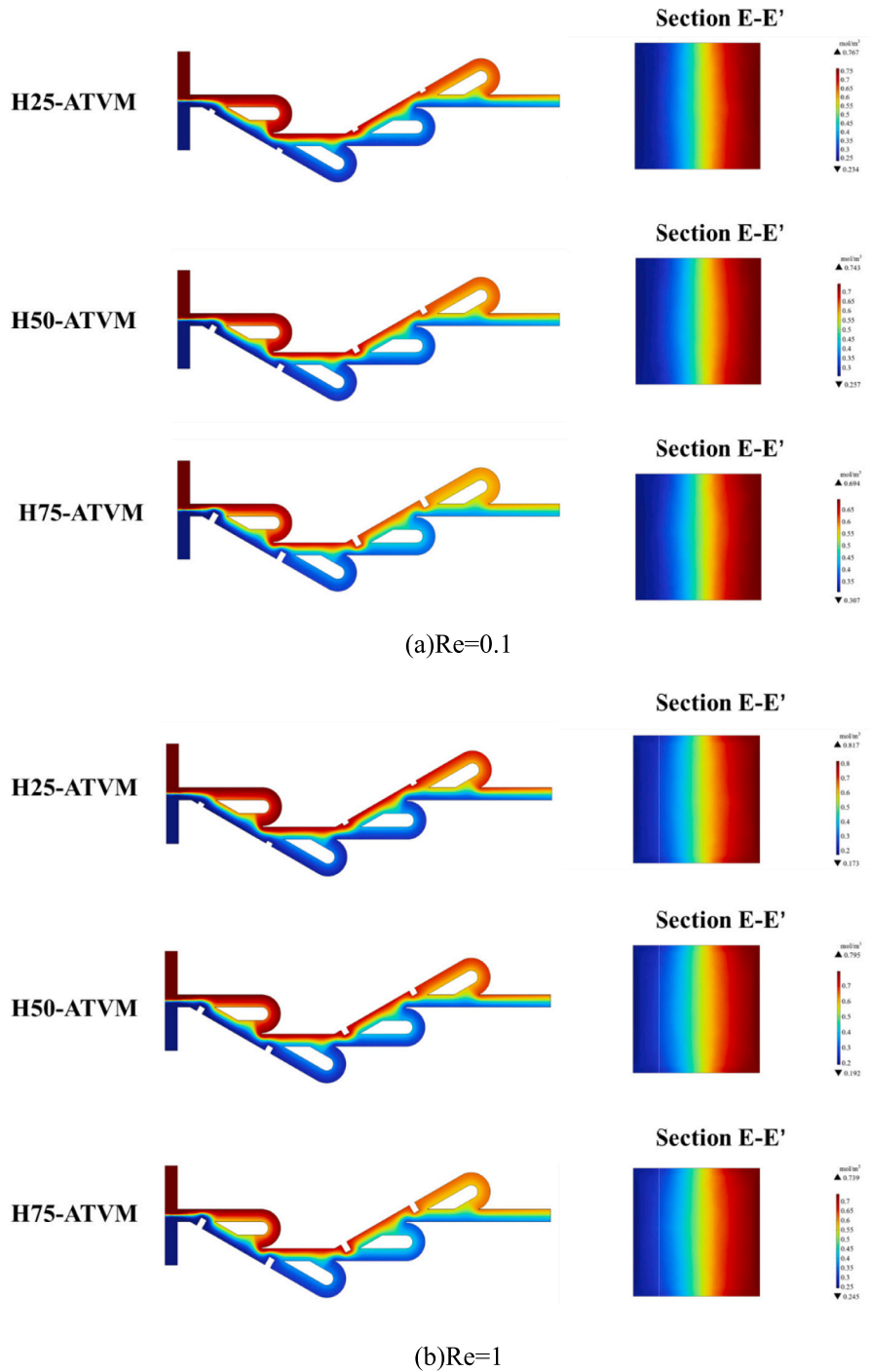


Fig. 9. Concentration field distribution at the horizontal cross section, outlet concentration distribution and outlet arrow distribution of H25-ATVM, H50-ATVM and H75-ATVM at different Re (a) $Re = 0.1$ (b) $Re = 1$ (c) $Re = 10$ (d) $Re = 100$.

addition of obstacle structures in the micromixer. However, those traditional methods will cause the problem of complex mixer structure. According to the research results of others, the micromixer with a baffle structure can significantly improve the mixing efficiency of the micromixer. Compared with other micromixers, the micromixer with a baffle structure can reduce the structural size and manufacturing cost of the micromixer [31]. Therefore, we proposed the method of adding baffles in the micromixer to study the mixing performance of the mixer. In this section, we proposed three baffle mixers with different structures, namely rectangular baffle asymmetric Tesla vortex micromixer (RB-

ATVM), triangular baffle asymmetric Tesla vortex micromixer (TB-ATVM) and square baffle asymmetric Tesla vortex micromixer (SB-ATVM). The geometric dimensions of these three baffle structures are shown in Fig. 5. The geometric dimensions of the rectangular baffle are. The geometric dimensions of the triangular baffle are. Through numerical simulation, we calculated the micromixer mixing efficiency, micromixer pressure drop and micromixer energy consumption curve of these three different baffle micromixers at different Re . We also compared these three baffle micromixers with different structures with the BM-ATVM.

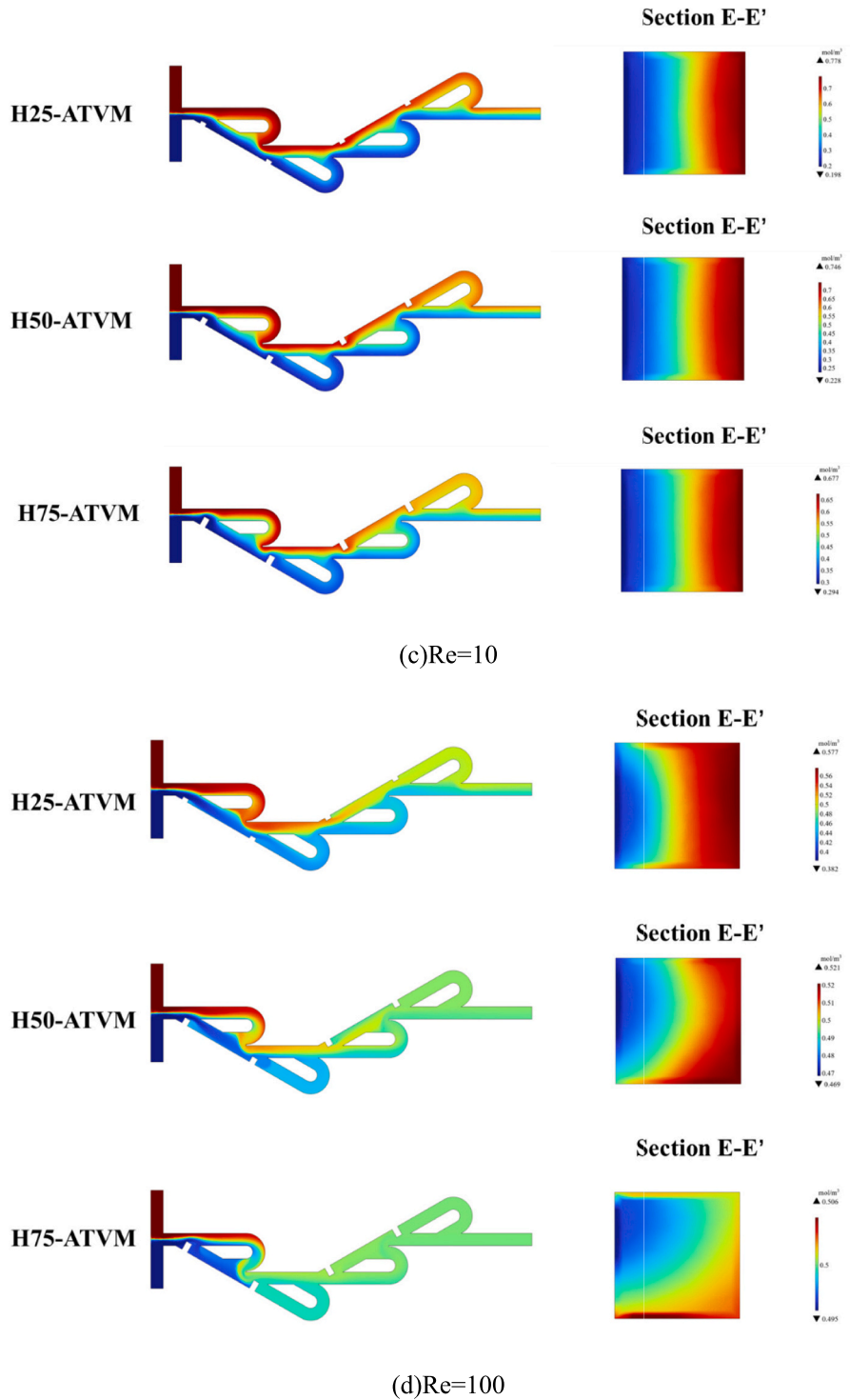


Fig. 9. (continued).

From Fig. 6, it can be found that the mixing efficiency of SB-ATVM is the micromixer with the highest mixing efficiency and pressure drop among the four micromixers. At the same time, it is also the micromixer with the lowest mixing energy consumption. As can be seen from Fig. 6 (a), the mixing efficiency of the four micromixers changes with Re and the change rate are basically the same. The mixing efficiency of the micromixers all show a trend of decreasing first and then increasing. This shows that when the baffle structure is added to the micromixer, the mixing efficiency of the micromixer can be effectively improved. And the mixing efficiency can be significantly improved without changing

the overall structural length of the micromixer. This greatly reduces the complexity of the micromixer and saves resources. As can be seen from Fig. 6 (b), when $Re = 0.1-10$, the pressure drop of the four micromixers is basically the same. However, when $Re = 10-100$, the pressure drop increase rate of the four micromixers is inconsistent. The pressure drop of SB-ATVM increases the fastest. And the pressure drop of SB-ATVM is the largest among the four when $Re = 100$. This shows that a large flow resistance is generated in SB-ATVM. This also shows that when the baffle structure changes, the effect of the pressure drop of the micromixer is particularly significant. As can be seen from Fig. 6 (c), the energy

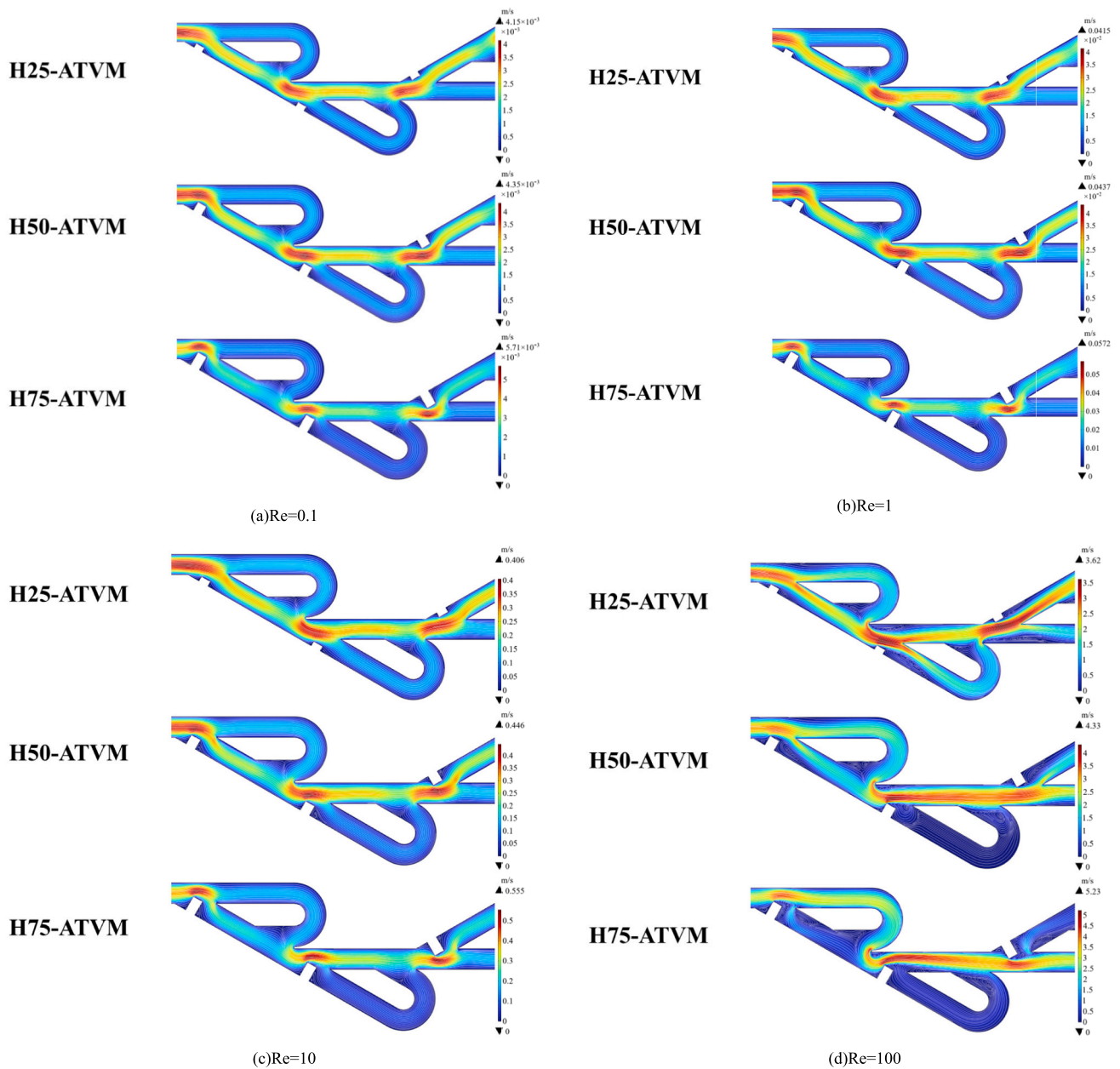


Fig. 10. Velocity distribution and streamline distribution at the horizontal section of H25-ATVM, H50-ATVM and H75-ATVM at different Re (a) $Re = 0.1$ (b) $Re = 1$ (c) $Re = 10$ (d) $Re = 100$.

consumption trends of the four micromixers are consistent. They all show a decrease with increasing Re . When the energy consumption of the four micromixers is $Re = 0.1$ – 10 , the energy consumption of SB-ATVM decreases the slowest. This shows that the mixing performance of SB-ATVM is superior to the other three. This shows that when $Re = 0.1$ – 10 , the square baffle structure in the micromixer will reduce the energy consumption of the micromixer.

As shown in Fig. 7, in order to analyze the influence of the baffle structure on the mixing process and mixing mechanism of the micromixer in more detail, we obtained the concentration field distribution, velocity streamline distribution, concentration distribution at the E-E' section and velocity arrow distribution diagram of the new asymmetric Tesla vortex micromixer with three different structural baffles at different Re . As can be seen from Fig. 7 (a), when $Re = 0.1$, a relatively obvious concentration stratification is generated at the horizontal section of the main channel in the three micromixers, and complete mixing cannot be achieved. This shows that when Re is small, the mixing

mechanism of the micromixer is mainly dominated by molecular diffusion. And no vortex is generated near the baffle, and there is almost no chaotic convection. When $Re = 1$, the mixing concentration of the three micromixers reaches the lowest. This is because the fluid velocity increases and the molecular diffusion effect decreases. But at this time, the chaotic convection effect is also almost small. Therefore, when $Re = 1$, the mixing efficiency of the three micromixers is the lowest. When $Re = 10$, it can be seen from Fig. 7 (c) that a certain vortex state is generated near the baffle structure and in the main channel. This shows that chaotic convection has begun to dominate fluid mixing at this time. However, since the chaotic convection effect has not been enhanced, the mixing efficiency has only been improved and has not been sharply enhanced. When $Re = 100$, compared with Fig. 4, it can be found that more vortices are generated near the baffles and in the main channels of the three baffle structure micromixers. This is because the baffle structure generates stable secondary vortices in the main channel of the micromixer. It is precisely because of more vortices that the mixing is

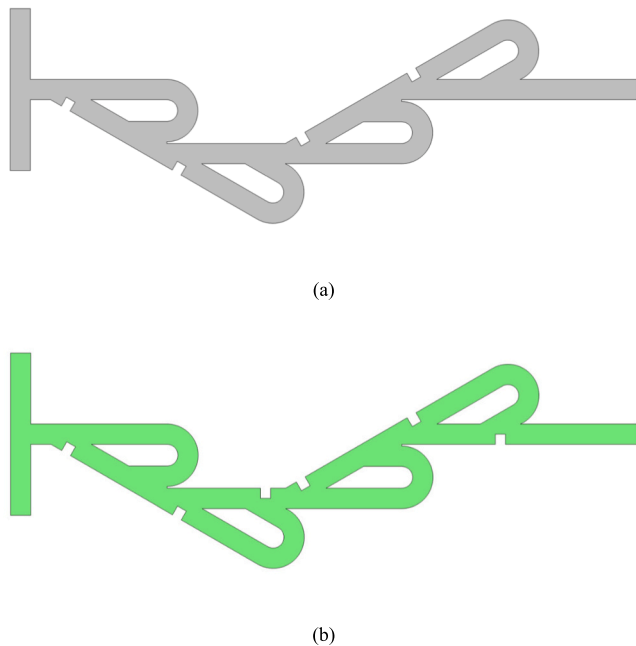


Fig. 11. Schematic diagram of the structure of BM-ATVM with different number square structural baffles (a) 4 square baffle, N4-ATVM (b) 6 square baffle, N6-ATVM.

enhanced. By comparing the mixing performance of three micromixers with baffles of different structures, it is obvious that the mixing performance of SB-ATVM is the best among the four. It can be clearly seen from Fig. 7 (d) that the secondary vortex state generated in SB-ATVM is more conducive to stretching and compressing the fluid. As a result, the chaotic convection effect is enhanced and the fluid mixing efficiency is enhanced.

The rectangular baffle ($50 \times 100 \mu\text{m}$) enhances the flow field instability through longitudinal large-scale shear layer separation, but its right-angle edge induces local backflow and causes additional pressure drop loss. The inclined surface of the triangular baffle (bottom edge $50 \mu\text{m}$) guides the production of directional secondary flow. Although it reduces the flow resistance (ΔP only increases by 12 %), the vortex intensity at the acute angle vertex is weak and the interface stretching efficiency is limited. The square baffle ($100 \times 100 \mu\text{m}$) forms a symmetrical Karman vortex street in the contraction section of the Tesla valve by virtue of its four-sided symmetrical geometric characteristics (Fig. 7d). Its periodically shed vortex pairs enhance the three-dimensional flow folding, which increases the concentration interface deformation rate to 2.3 times that of the rectangular baffle. Quantitative analysis shows that the square structure achieves the best result when $Re = 100$, and the stable vortex induced by its right-angle edges shortens the mixing path length, ultimately achieving the coordinated optimization of 99.42 % mixing efficiency and 3.74×10^{-5} energy consumption index.

3.3. Effect of baffle heights on mixing performance of asymmetric tesla vortex micromixer

In order to study the influence of other geometric dimensions of the baffle on the mixing performance of the micromixer, in this section, based on the BM-ATVM model, we adopted a square baffle structure and designed three different baffle heights H3 ($20 \mu\text{m}$, $50 \mu\text{m}$ and $75 \mu\text{m}$). These three micromixers with different baffle heights are named H25-ATVM, H50-ATVMHE and H75-ATVM. As shown in Fig. 8, we obtained the micromixer mixing efficiency curve, micromixer pressure drop curve and micromixer energy consumption curve of the three

micromixers with different baffle heights at different Re . It can be seen from Fig. 8 (a) that the mixing efficiency of the three micromixers shows a consistent trend with Re . Fig. 8 (a) shows that the mixing efficiency of H75-ATVM is the highest. It can be seen from Fig. 8 (a) that when $Re = 0.1$, after the baffle height increases from $25 \mu\text{m}$ to $75 \mu\text{m}$, the mixing efficiency of the micromixer increases significantly from 0.65 to 0.75. However, as Re increases to 100, the mixing efficiency of H50-ATVM and H75-ATVM is not much different. However, as can be seen from Fig. 8 (b), when $Re = 100$, the pressure drop of H50-ATVM and H75-ATVM is very different, reaching 26,629.14 Pa. This also shows that H75-ATVM has a small improvement in mixing efficiency, but loses more pressure. Fig. 8 (c) shows that the mixing energy consumption of the three micromixers decreases with increasing Re . The energy consumption of the three is almost the same at $Re = 100$. At low Re , the mixing energy consumption of H75-ATVM is lower.

In order to further analyze the huge differences in mixing efficiency, pressure drop and mixing energy consumption among the three micromixers, we plotted the concentration distribution of the horizontal section of the main channel of the three micromixers, the concentration distribution at the E-E' section, the local velocity distribution of the main channel and the velocity streamline distribution as shown in Fig. 9 and Fig. 10. It can be clearly seen from Fig. 9 that when $Re = 0.1$, the three micromixers failed to achieve sufficient mixing. The concentration stratification can be clearly seen at the horizontal section of the main channel of the three micromixers. And from the concentration distribution at the E-E' section, it can be seen that the maximum and minimum concentration difference of H75-ATVM is the smallest. This shows that the mixing efficiency of H75-ATVM is higher than that of the other two. As Re increases, we can see from the concentration difference of the E-E' section that the concentration difference of the three mixers is decreasing. This is because as the height of the baffle increases, the chaotic convection effect generated in the microchannel is stronger, accelerating the mixing. As Re increases to 100, the concentration stratification in the three micromixers gradually presents a curved and folded state. It is worth noting that the H75-ATVM fluid flow is hindered due to the increase in the local baffle, which intensifies the bending and folding path and flow state of the fluid. This enhances the mixing efficiency of the micromixer. As can be seen from Fig. 9 (d), at this time, the concentration difference of H75-ATVM at the E-E' section is the smallest among the three. We can obtain that when $Re = 0.1$, 1, 10, 100, the concentration difference of H75-ATVM at the E-E' section is 0.387 mol/m^3 , 0.494 mol/m^3 , 0.383 mol/m^3 and 0.011 mol/m^3 respectively.

Fig. 10 describes the local velocity distribution and velocity streamline distribution at the horizontal section of the main channel of the micromixer for three different micromixers at different Re . In order to better quantify the vortex intensity, we use velocity streamlines to compare the vortex intensity. When the vortex intensity is large, denser velocity streamlines will be generated in the microchannel. From Fig. 10 (a), it can be seen that when $Re = 0.1$, the velocity streamline distribution is very uniform. The fluid does not generate vortices when passing through the baffle. This also proves that the mixing of the micromixer at this time mainly relies on molecular diffusion. As Re increases, it can be clearly seen in Fig. 10 (b ~ d) that when the fluid flows through the baffle, secondary flows with different shapes and intensities will be generated. Thereby, the fluid in the micromixer is prompted to break the laminar flow state. As Re increases to 100, the secondary vortex generated in the downstream channel after the fluid flows through the baffle continues to expand downstream. From Fig. 10 (d), it can be seen that there are differences in the vortex intensity and vortex area generated in the three micromixers. The vortex intensity generated near the baffle of H75-ATVM is the strongest and the vortex morphology area generated downstream is larger. Therefore, this is beneficial for H75-ATVM to improve the chaotic convection effect in the mixing process, thereby enhancing the mixing.

The formation of the second vortex is actually the key control mechanism for enhanced mixing. As shown in Fig. 10(d), the

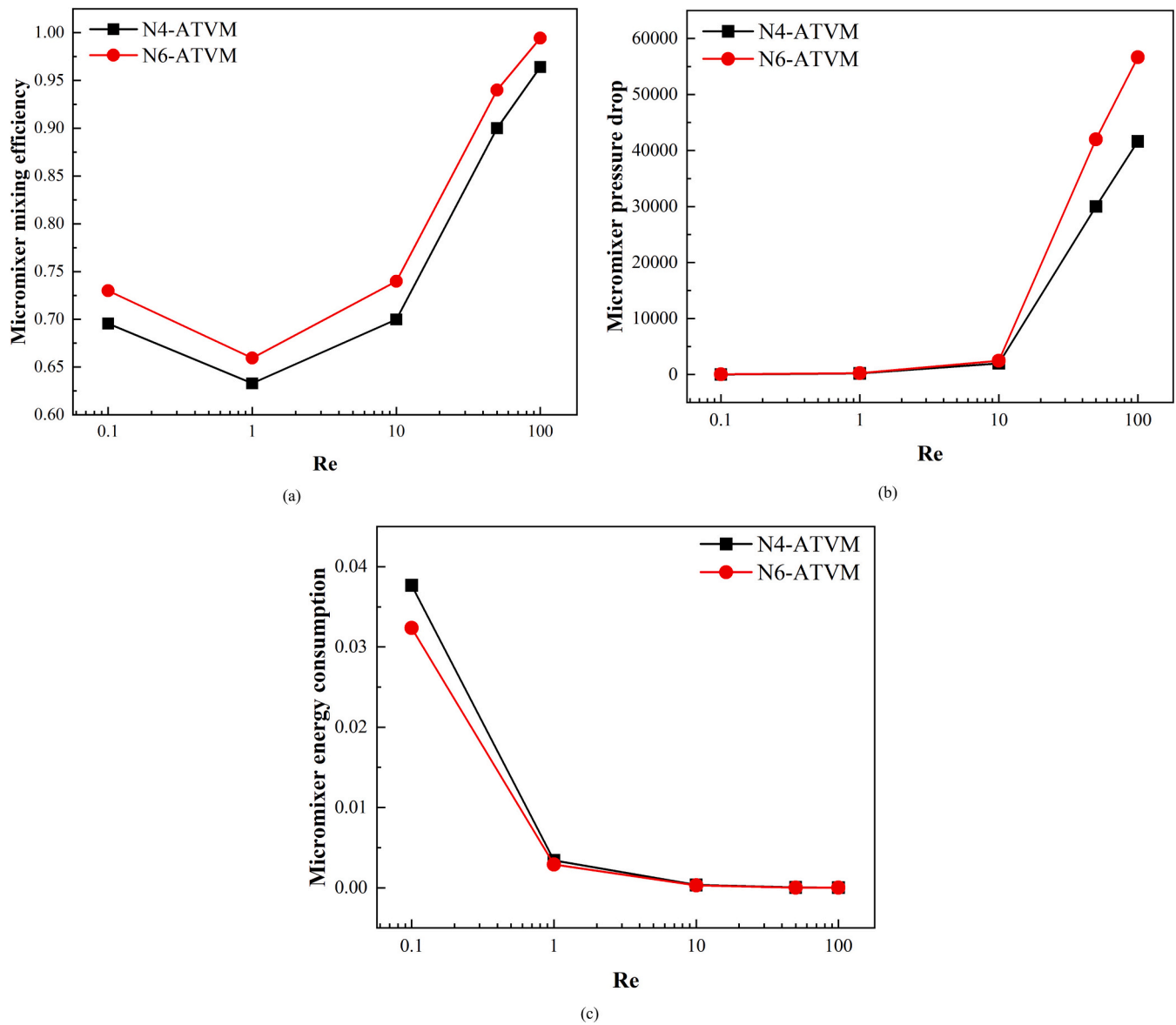


Fig. 12. Mixing efficiency curve, pressure drop curve and mixing energy consumption curve of N4-ATVM and N6-ATVM at different Re (a) Mixing efficiency (b) Pressure drop (c) Mixing energy consumption.

symmetrical secondary vortex formed downstream of the baffle partially reduces the axial flow velocity, but significantly improves the mixing performance in the following aspects. The periodic rotation of the vortex prolongs the retention time of fluid particles and increases the molecular diffusion time by about 40 %, overcoming the limitation of insufficient diffusion efficiency at low Re numbers. The centrifugal effect generated by the vortex core drives the radial stretching of the fluid interface, greatly increasing the contact area of heterogeneous fluids. The shear-folding coupling of the vortex destroys the concentration boundary layer and increases the concentration gradient by 3.2 times. This design converts the “obstruction effect” into a kinetic advantage for enhanced mixing by precisely controlling the vortex scale and spatial distribution.

3.4. Effect of baffle numbers on mixing performance of asymmetric tesla vortex micromixer

According to the research in the previous section, simply increasing the height of the baffles to improve the mixing performance will result in huge pressure drop losses. While we understand how the height of the baffles affects the mixing performance of the micromixer, we study the

effect of different numbers of baffles on the mixing performance of the micromixer in this section. We designed micromixers with four and six square baffles in the microchannel, respectively, and they were named N4-ATVM and N6-ATVM (as shown in Fig. 11). Increasing the number of baffles will not cause the micromixer structure to grow and become complicated. Through numerical simulation, we calculated the micromixer mixing efficiency curves, micromixer pressure drop curves and micromixer energy consumption curves of N4-ATVM and N6-ATVM at different Re as shown in Fig. 12. As can be seen from Fig. 12 (a), the mixing efficiency change trends of N4-ATVM and N6-ATVM are consistent. And their mixing efficiency difference remains basically stable. When $Re = 100$, the mixing efficiency of N6-ATVM reaches 99.42 %, achieving full mixing of the fluid. This shows that the mixing efficiency of the fluid is higher after flowing through more baffles. It is precisely because of more baffles that more secondary flows are generated in the micromixer, thereby enhancing mixing. As can be seen from Fig. 12 (b), when Re is less than 10, the pressure drops of N4-ATVM and N6-ATVM are basically the same. When $Re = 100$, there is a difference in the pressure drops of N4 and N6. This is because the fluid flows through more baffles, resulting in greater flow resistance. Compared with the

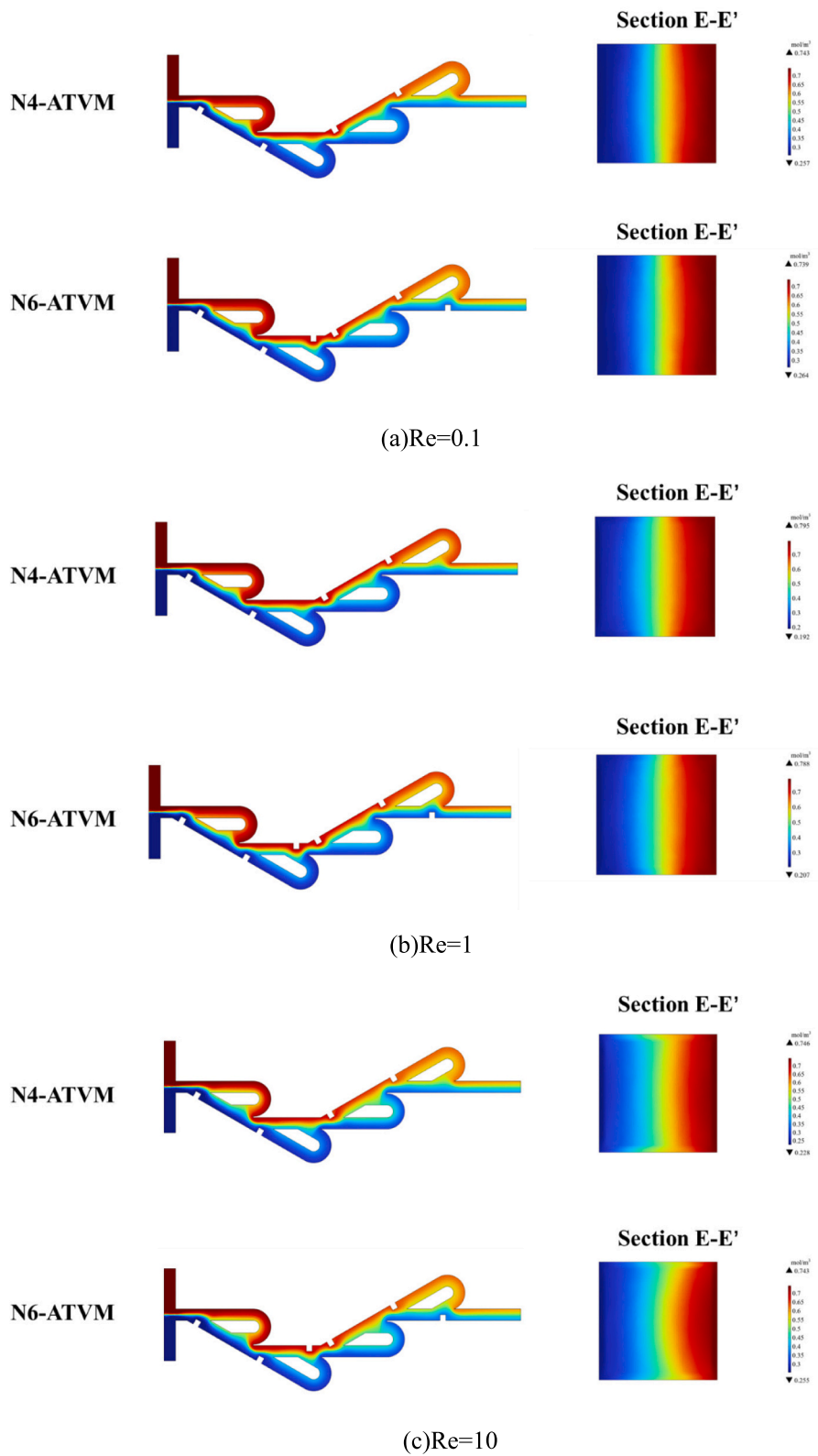


Fig. 13. Concentration field distribution at the horizontal cross section, outlet concentration distribution and outlet arrow distribution of N4-ATVM and N6-ATVM at different Re (a) $Re = 0.1$ (b) $Re = 1$ (c) $Re = 10$ (d) $Re = 100$.

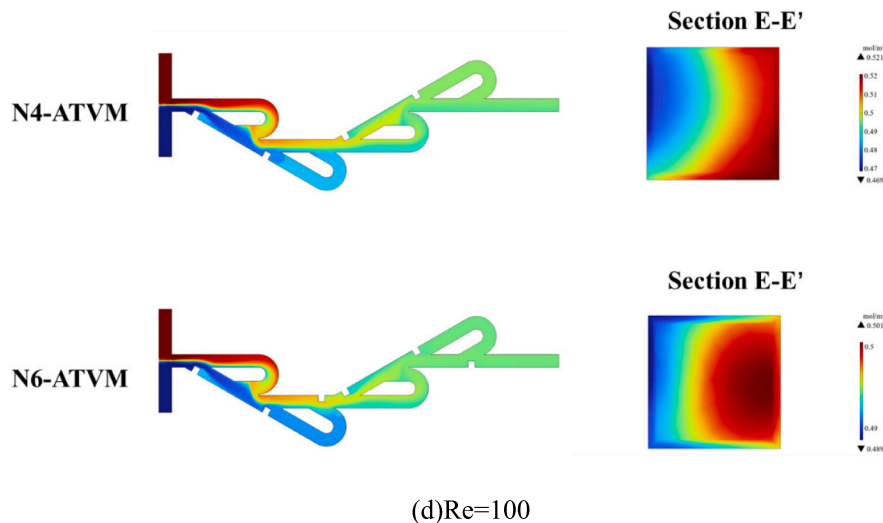


Fig. 13. (continued).

pressure drop caused by increasing the baffle height, the pressure drop caused by using more baffles is lower. This also achieves highly uniform mixing with low pressure drop. As can be seen from Fig. 12 (c), there is a difference in the energy consumption of the micromixers of N4-ATVM and N6-ATVM at lower Re numbers. This shows that the energy consumption required for N4-ATVM to achieve complete mixing is higher than that of N6-ATVM.

In order to further explain the reasons for the above research results, the concentration distribution at the horizontal section of the main channel of the two micromixers, the concentration distribution at the E-E' section, the local velocity distribution of the main channel and the velocity streamline distribution are shown in Fig. 13 and Fig. 14. As can be seen from Fig. 13 (a), when the number of baffles increases, the concentration stratification length of the fluid increases at $Re = 0.1$. This is because the increase in the number of baffles will increase the fluid flow distance. As a result, the fluid undergoes a longer stretching and compression path, resulting in enhanced mixing. And the difference between the maximum concentration and the minimum concentration of N6-ATVM at the E-E' section is the smallest. However, at this time, the fluid concentration difference is large, and the fluid fails to achieve full mixing. As Re continues to increase to 100, N6-ATVM achieves complete mixing, and the concentration difference at the E-E' section is 0.002 mol/m^3 . When the number of baffles increases, the local mixing inside the micromixer is enhanced, thereby improving the mixing efficiency of the micromixer. From the comparison of the E-E' section concentration difference under different Re , the concentration difference of N6 is the smallest. When $Re = 0.1, 1, 10, 100$, the concentration difference of N6-ATVM is $0.475 \text{ mol/m}^3, 0.581 \text{ mol/m}^3, 0.488 \text{ mol/m}^3$ and 0.002 mol/m^3 respectively. Fig. 14 shows the local flow field state of the micromixer at different Re . When $Re = 0.1$, it can be seen that the streamline distribution of N4-ATVM and N6-ATVM is very uniform, and no vortex is generated. When $Re = 100$, it can be seen that multiple vortices are generated near the baffle and in the downstream area of the baffle. Since N6-ATVM has more baffles, the number and total area of vortices generated by N6-ATVM are more than those of N4-ATVM. This shows that as the number of baffles increases, the number and area of vortices generated in the micromixer increase, thereby enhancing fluid mixing. By analyzing the effects of baffle structure, baffle height and number of baffles on the mixing performance of the micromixer, we determined that the best mixing performance of the micromixer can be achieved when 6 square baffles with a height of $50 \mu\text{m}$ are used. By comparing with previous studies, the baffle structure micromixer developed and designed in this study shows the advantages of simpler structure, low energy consumption and efficient mixing.

4. Conclusion

Through numerical simulation and structural optimization, this study successfully verified the innovative value of asymmetric Tesla vortex micromixer in microscale fluid mixing. The research results show that the mixer achieves efficient mixing under low Reynolds number conditions through spontaneous flow separation mechanism, and its performance is significantly better than that of traditional designs. The adoption of square baffle structure achieves a good balance between mixing efficiency and energy consumption control, showing excellent application potential. The specific conclusions are as follows:

(1) The mixing efficiency of BM-ATVM at low Reynolds number decreases first and then increases. When $Re = 100$, the mixing efficiency of BM-ATVM reaches 85.57 %. The Tesla valve structure breaks the laminar flow through vortex, significantly improving the mixing performance, but the pressure drop increases significantly with the increase of Re .

(2) SB-ATVM has the highest mixing efficiency and the lowest energy consumption. The stable secondary vortex of SB-ATVM significantly enhances chaotic convection. The baffle structure achieves more efficient mixing in the low Reynolds number range by changing the fluid motion trajectory.

(3) The H75-ATVM with a baffle height of $75 \mu\text{m}$ has the best efficiency at low Re , but the difference with the $50 \mu\text{m}$ baffle is reduced at high Re . Although the increase in baffle height enhances the chaotic effect, it will lead to increased pressure loss, and the efficiency and energy consumption need to be weighed.

(4) The efficiency of N6-ATVM reaches 99.42 % at $Re = 100$, and the energy consumption required for complete mixing is lower. Increasing the number of baffles can increase the number and area of vortices and significantly enhance the fluid mixing effect. The performance is best when the number of optimized baffles is 6.

The results of this study not only deepen the understanding of the microscale mixing mechanism, but also open up a new path for the application of microfluidic technology in the fields of biochemical analysis and drug synthesis. Subsequent research can further explore the optimization of mixer performance under complex multiphase flow conditions.

The asymmetric Tesla vortex structure (ATVM) designed in this study can significantly improve the performance of key scenarios. In the synthesis of nanomedicine, the 6-baffle configuration ensures the stability of bioactive substances through low-shear vortex. In the field of organ chip perfusion, low-flow mixing with $Re < 10$ meets the cell shear sensitivity threshold. For instant diagnostic equipment, high energy

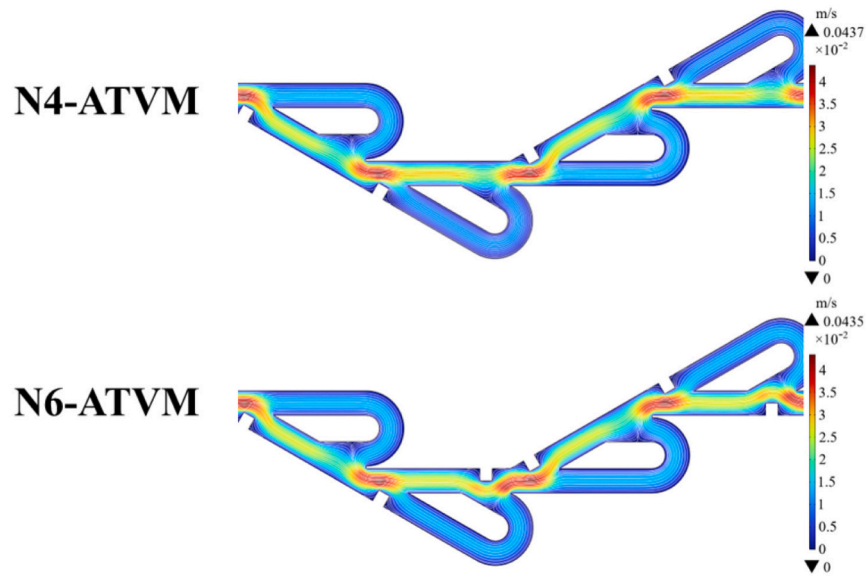
(a) $Re = 0.1$ (b) $Re = 1$

Fig. 14. Velocity distribution and streamline distribution at the horizontal section of N4-ATVM and N6-ATVM at different Re (a) $Re = 0.1$ (b) $Re = 1$ (c) $Re = 10$ (d) $Re = 100$.

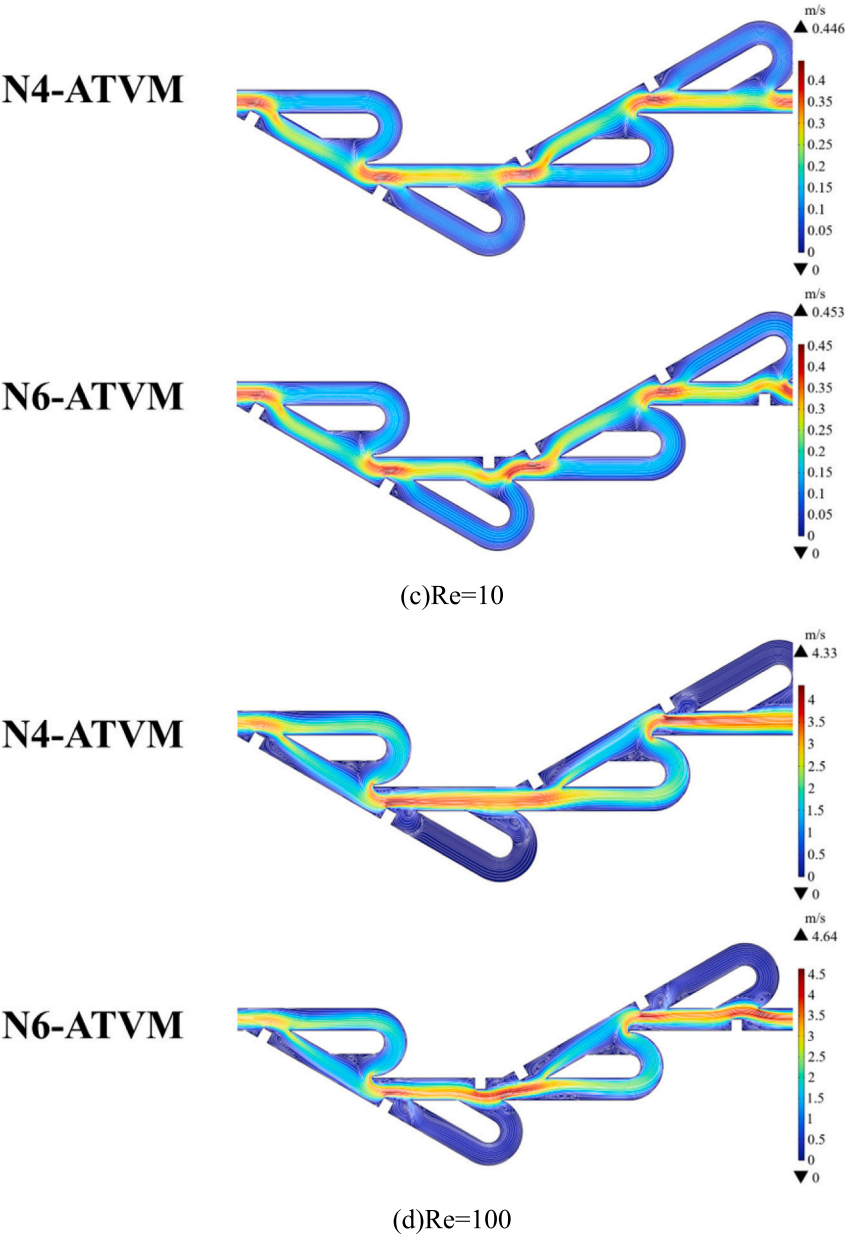


Fig. 14. (continued).

efficiency ratio adaptive micropump drive constraints ensure the rapid response consistency of immunochromatographic reagents.

CRediT authorship contribution statement

Pu Liu: Writing – review & editing, Writing – original draft, Validation, Supervision, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Yiheng Yue:** Validation, Software, Resources, Methodology, Investigation. **Yu Guo:** Visualization, Software, Project administration. **Wei Yang:** Investigation, Funding acquisition. **Lili Zang:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization.

Consent to participate

Not applicable.

Consent for publication

Not applicable.

Ethics approval

Not applicable.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.icheatmasstransfer.2025.109663>.

Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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